18 UNEXPECTED WAYS 3D PRINTERS ARE BEING USED

With 3D printing getting more accessible over the years, we are beginning to see some outlandish creations make their way into the market. These are 18 unexpected ways 3D printers are being used.

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3D printing, to the average Joe, may seem like a relatively new innovation but the concept was actually first formulated and made commercially available in the 1980s. Since then the technology has progressed meteorically and is being used widely across the world. Markets and Markets has reported that the 3D printing industry's worth will be nearly $33 billion USD in 2023. Businesses and consumers continually purchase these printers since the possibilities for their use seem to be endless. Here are eighteen of the most unexpected ways in which 3D printers are being used today.

01 HOUSING

Apart from conceptualising and designing homes using 3D printers, the actual creation of houses is also possible with this technology. The 3D printers used to

3D printed homes are seemingly more affordable than regular ones

accomplish this are portable and are able to build homes in a large number of shapes and sizes. Additionally, the layered structure of these houses allow for easy installation of plumbing, electrical lines and insulation. It is also increasingly being seen as one of the methods to obtain affordable housing. An American startup, ICON, unveiled a 650-square-foot house that was 3D printed out of concrete in under 24 hours in 2018. The house costs a mere $4000 and successfully complies with local building standards in the US.

02 BIONIC BODY PARTS AND PROSTHESES

3D printers are being used to create bionic body parts such as hands, limbs and even eyes now. Currently, the bionic eye will not be able to restore sight but can make sure that the person's facial bone structure grows properly around the bionic eye, thereby giving him a more normal appearance. Restored sight with a 3D printed bionic eye is a very real possibility in the future and the groundwork is already being laid for it. 3D printed prostheses are extremely efficient since they are customised specially for each patient. Create Orthotics and Prosthetics was the first company to build medical-grade 3D printed limbs that could be customised by the patients. Enabling the Future is a nonprofit organisation that allows paraplegic children to connect with volunteers who can make custom 3D printed hands for them.

03 FOOD

Some companies work with 3D printers in the food industry to print intricate and beautiful-looking food items such as desserts and pizzas. Others have begun delving into this area to possibly print complete meals from a dispenser unit. The latter can help feed thousands in rural areas that do not have means to healthy, nutritional meals. Companies working with large scale 3D food printers are working towards creating complete meals within a few minutes with a couple of ingredients. Another application of 3D food printing is making smooth food for elderly people, who cannot chew, appear like real, solid food. So instead of getting some unappetising puree, elders can eat food that is smooth and easy-to-chew but in 'normal food shapes'. This is made by reshaping pureed food to normal shapes in 3D printed silicone molds.

03 SCULPTURES

3D printed sculptures range from absolute massive to miniscule pieces of art. They are made from all kinds of materials like plastic, paper, bronze, and other metals. Since the technology has limitless design possibilities, it allows artists